

# Technical Document #830: Software Engineering - Comprehensive Overview

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Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

## Key Takeaways:

- Refactoring improves code structure without changing behavior.
- Continuous deployment (CD) automatically deploys code changes to production.
- Continuous integration (CI) automatically builds and tests code changes.